

Title: Resource-Aware Quantum Routing in Quantum Networks

Objective: Efficient Path Selection and Resource Management for Quantum Network Simulators

6-Week Internship Roadmap

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Project Synopsis

This project develops a simulator to model quantum communication networks and QKD-based systems. Unlike classical network routing, where the shortest or fastest path is often sufficient, routing in quantum networks must account for fragile quantum resources such as memory lifetime, fidelity loss, probabilistic link generation, detector imperfections, and entanglement swapping success probability.

Participants will study current research on quantum routing, entanglement distribution, trusted-node QKD networks, repeater-based routing. They will design routing algorithms that estimate the most efficient pathway between a source and target node using node, edge, detector, and swapping parameters.

The final goal is to create a prototype routing framework that can compare multiple candidate paths and select the most suitable route based on success probability, expected latency, final fidelity, memory availability, detector performance, and resource contention.

Week-by-Week Plan

Week 1: Foundations and Literature Review

- Quantum Network Fundamentals:
 - Study the difference between classical routing and quantum routing.
 - Understand trusted-node QKD networks, repeater-based networks, and entanglement-based routing.
 - Review QNTSim architecture, including QKDNode, QuantumRouter, BSMNode, channels, detectors, and protocol stack structure.

- Literature Reading:

- Entanglement routing in quantum networks.
- Fidelity-guaranteed quantum routing algorithms.
- Opportunistic routing in quantum networks.
- Multi-user entanglement distribution and multipath routing.
- SeQUeNCe-style resource-management architecture.

- Outcome:

- Prepare a short literature summary explaining key routing challenges: probabilistic link generation, memory decoherence, detector noise, classical delay, and swapping failures.

Week 2: Routing Metrics and Cost-Model Design

- Parameter Identification:

- Node parameters: memory size, operation frequency, fidelity, coherence time, and efficiency.
- Edge parameters: attenuation, distance, quantum delay, and classical delay.
- Detector parameters: count rate, dark count rate, and detector efficiency.
- Swapping parameters: Bell-state measurement success probability, swapping fidelity, and swapping operation time.

- Cost-Model Formulation:

- Define link success probability using attenuation, distance, detector efficiency, and node efficiency.
- Estimate expected link-generation time.
- Model memory survival using coherence time.
- Estimate route-level success probability, final fidelity, expected latency, and resource cost.

- Outcome:

- Develop a formal route-scoring model that ranks candidate paths using success probability, fidelity, latency, memory usage, and bottleneck penalties.

Week 3: Baseline Routing Implementation

- Algorithm Implementation:

- Implement shortest-path routing as a baseline.
- Implement loss-aware routing using distance and attenuation.
- Implement success-probability-aware routing using node, edge, and detector parameters.

- Candidate Path Generation:

- Generate multiple source-to-target paths for a given topology.
- Compute per-path metrics such as hop count, expected transmission probability, classical delay, and estimated link-generation time.

- Validation:

- Test algorithms on small network topologies of 5-7 nodes.
- Compare cases where shortest path is not the best quantum path because of high attenuation, poor detector efficiency, or memory limitations.

- Outcome:

- Produce a working baseline routing prototype and a comparison table of candidate routes.

Week 4: Fidelity-Aware and Resource-Aware Routing

- Fidelity-Constrained Routing:

- Add final-fidelity estimation for each candidate route.
- Reject paths that fall below a target fidelity threshold.
- Include fidelity degradation due to memory waiting time and swapping operations.

- Resource-Aware Routing:

- Include memory availability, detector load, and swapping-resource availability in routing decisions.

- Add bottleneck detection for shared intermediate nodes.
- Model resource contention when multiple source-target sessions run simultaneously.

- Multi-Request Routing:

- Study simultaneous routing requests, such as key sharing between node pairs (1,6) and (5,2) through shared trusted nodes.
- Propose scheduling policies such as FIFO, priority-based, fair-share, max-throughput, and max-fidelity scheduling.

- Outcome:

- Implement a resource-aware routing layer that can rank, accept, reject, or defer routing requests based on current network resource state.

Week 5: ResourceManager Design and Visualization

- ResourceManager Redesign:

- Design node resource tables tracking memory availability, coherence time, efficiency, and active sessions.
- Design edge resource tables tracking distance, attenuation, delay, and current load.
- Design detector resource tables tracking efficiency, dark count, count rate, and busy time.
- Design swapping resource tables tracking BSM success probability, operation time, and fidelity loss.
- Design routing session tables with session ID, source, target, selected path, reserved resources, and status.

- Visualization Design:

- Candidate path ranking panel.
- Resource heatmap for memory, detector load, and bottleneck nodes.
- Route execution timeline showing link generation, heralding, memory reservation, swapping, and final path creation.
- Fidelity decay timeline showing how stored quantum states lose quality over time.
- Rejected-route explanation panel showing why a route failed.

- Outcome:

- Prepare implementation-ready ResourceManager specifications and UI/visualization schemas for QNTSim.

Week 6: Benchmarking, Documentation, and Finalization

- Benchmarking:

- Compare shortest-path, loss-aware, fidelity-aware, and resource-aware routing algorithms.
- Benchmark routing performance on multiple network topologies.
- Measure selected path quality, expected success probability, final fidelity, latency, and resource usage.

- Documentation:

- Write a technical report explaining routing models, algorithms, assumptions, and results.
- Prepare developer documentation for ResourceManager integration.
- Create example topologies and sample configuration files.

- Final Presentation:

- Prepare presentation slides summarizing motivation, methods, routing algorithms, benchmarks, and future work.
- Identify future extensions such as purification-aware routing, opportunistic routing, and dynamic rerouting.

- Outcome:

- Final prototype, benchmark report, documentation, and presentation.

Mentorship and Workflow

- Weekly one-hour video calls for technical guidance and progress review.
- Code reviews through GitHub pull requests.
- Shared Kanban board for task tracking.

- Weekly progress updates with short written summaries.
- Final review session for code, documentation, and presentation.

Expected Deliverables

- A literature review summary on quantum routing, entanglement routing, resource-aware routing, and multi-user quantum network routing.
- A formal routing-cost model using node, edge, detector, and swapping parameters.
- Prototype routing algorithms including most efficient-path, loss-aware, fidelity-aware, and resource-aware routing.
- A ResourceManager design proposal for QNTSim, including resource tables and session-management logic.
- Benchmark reports comparing routing strategies across multiple network topologies.
- Visualization schemas for candidate path ranking, resource heatmaps, route timelines, and fidelity decay.
- Example topology configurations for testing 5-7 node networks.
- Final presentation slides and a technical report.

Contact

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